

3.2)

$V =$ prostor možných reálných
funkcí $\cap D_0 = [1, 5]$

$$\langle f(x), g(x) \rangle = \int_1^5 f(x) g(x) dx$$

a) ON báze $\text{span}\{1, x, x^2\}$

1) $v_1 = 1$

$$\|v_1\| = \sqrt{\int_1^5 1^2 dx} = \sqrt{[x]_1^5} = \sqrt{4} = 2$$

$$u_1 = \frac{v_1}{\|v_1\|} = \underline{\underline{\frac{1}{2}}}$$

2) $v_2 = x$

$v_2 = 0$ je projekce v_2 na $\text{span}\{u_1\}$

$$\begin{aligned}
 m_1 &= \langle m_1, m_2 \rangle \quad m_1 = \int_1^5 x \cdot \frac{1}{2} dx \cdot \frac{1}{2} = \\
 &= \left[\frac{x^2}{4} \right]_1^5 \cdot \frac{1}{2} = \left(\frac{25}{4} - \frac{1}{4} \right) \cdot \frac{1}{2} = \frac{6}{2} = 3
 \end{aligned}$$

$$m_2 - m_1 = x - 3$$

$$\begin{aligned}
 \|m_2 - m_1\| &= \sqrt{\int_1^5 (x-3)^2 dx} = \sqrt{\int_1^5 x^2 - 6x + 9 dx} \\
 &= \sqrt{\left[\frac{x^3}{3} - 3x^2 + 9x \right]_1^5} = \sqrt{\frac{125}{3} - 75 + 45 - \left(\frac{1}{3} - 3 + 9 \right)} \\
 &= \sqrt{\frac{124}{3} - 30 - 6} = \sqrt{49 - 36 + \frac{1}{3}} = \sqrt{\frac{15+1}{3}} = \frac{4\sqrt{3}}{3}
 \end{aligned}$$

$$m_2 = \frac{m_2 - m_1}{\|m_2 - m_1\|} = \frac{x-3}{\frac{4\sqrt{3}}{3}} = \frac{\sqrt{3}}{4} \cdot (x-3)$$

$$3) m_3 = x^2$$

$\psi_2 = 0$ majerkel ψ_3 ora $\psi_{norm}(\psi_1, \psi_2)$

$$\psi_2 = \langle \psi_1 | \psi_3 \rangle \psi_1 + \langle \psi_2 | \psi_3 \rangle \psi_2 =$$

$$= \int_1^5 \frac{x^2}{2} dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3}}{4} (x-3)x^2 dx \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= \left[\frac{x^3}{6} \right]_1^5 \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3}}{4} x^3 - \frac{3\sqrt{3}}{4} x^2 dx \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= \left(\frac{125}{6} - \frac{1}{6} \right) \cdot \frac{1}{2} + \left[\frac{\sqrt{3}x^4}{16} - \frac{\sqrt{3}x^3}{4} \right]_1^5 \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= \frac{124}{6} \cdot \frac{1}{2} + \left[\frac{\sqrt{3}}{4} x^3 \left(\frac{x-1}{4} \right) \right]_1^5 \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= \frac{71}{3} + \left(\frac{\sqrt{3}}{4} \cdot \frac{125}{4} - \frac{\sqrt{3}}{4} \left(-\frac{3}{4} \right) \right) \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= \frac{71}{3} + \frac{\sqrt{3}}{16} \cdot 128 \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= \frac{31}{3} + 8\sqrt{3} \cdot \frac{\sqrt{3}}{4} (x-3) = \frac{31}{3} + 6(x-3) =$$

$$= \frac{31}{3} - \frac{54}{3} + 6x = -\frac{23}{3} + 6x$$

$$M_3 - M_2 = x^2 - 6x + \frac{23}{3}$$

$$\begin{aligned} \|M_3 - M_2\| &= \sqrt{\int_1^5 \left(x^2 - 6x + \frac{23}{3}\right)^2 dx} = \\ &= \frac{1}{3} \sqrt{\int_1^5 (7x^2 - 18x + 23)^2 dx} = \frac{1}{3} \sqrt{\frac{256}{5}} = \frac{1}{3} \frac{16}{\sqrt{5}} \end{aligned}$$

$$\begin{aligned} M_3 &= \frac{M_3 - M_2}{\|M_3 - M_2\|} = \frac{x^2 - 6x + \frac{23}{3}}{\frac{1}{3} \frac{16}{\sqrt{5}}} = \\ &= (7x^2 - 18x + 23) \cdot \frac{\sqrt{5}}{16} \end{aligned}$$

Or béaze: $\left(\frac{1}{2} \mid \frac{\sqrt{3}}{4} (x-3) \mid \frac{\sqrt{5}}{16} (7x^2 - 18x + 23) \right)$

b) ON možeme sim

m = ON možeme sim a na

Ymame m_1, m_2, m_3

m_1, m_2, m_3 tvoří ON bázi,
vyjádříme tedy:

$$m = \langle m_1, \sin x \rangle m_1 + \langle m_2, \sin x \rangle m_2 + \langle m_3, \sin x \rangle m_3$$

$$\langle m_1, \sin x \rangle = \int_1^5 \frac{\sin x}{2} dx = \left[-\frac{\cos x}{2} \right]_1^5 = \underline{\underline{\approx 0,128}}$$

$$\langle m_2, \sin x \rangle = \int_1^4 \frac{\sqrt{3}}{4} (x-3) \cdot \sin x dx \approx \underline{\underline{-1,493}}$$

$$\langle m_3, \sin x \rangle = \int_1^5 \frac{\sqrt{5}}{16} (3x^2 - 12x + 23) \cdot \sin x dx =$$

$$\lambda = 0,125$$

$$0,064 - 1,537$$

$$m \approx 0,128 \cdot \frac{1}{2} - 1,493 \cdot \frac{\sqrt{3}}{4} (x-3) -$$

$$- 0,125 \cdot \frac{\sqrt{5}}{16} (3x^2 - 18x + 27)$$

$$\approx -\frac{\sqrt{5} \cdot 3}{128} x^2 + \frac{\sqrt{5} \cdot 18}{10 \cdot 8} x - \frac{1,493 \cdot \sqrt{3}}{4} x -$$

$$- \frac{\sqrt{5}}{128} \cdot 27 + \frac{3\sqrt{3}}{4} \cdot 1,493 + 0,064 \approx$$

$$\approx -0,052x^2 - 0,332x + 1,544$$

$$3) || \lim x + 0,052x^2 + 0,332x - 1,544 || =$$

$$= \sqrt{\int_1^5 (\lim x + 0,052x^2 + 0,332x - 1,544)^2 dx}$$

$$= \underline{\underline{0,336}}$$